

MKM-EF-001 — Test Results

Component: Event Frequency Model — MKM-EF-001 (governance record v1.1.0) **Scope:** Unit and property-wiring test evidence for the frequency layer (`src/models/frequency/`) and its integration into gauge, property and commercial PRS pricing. **Result:** 325 tests passed, 0 failed · 99.67% line coverage (908 statements, 3 missed) · Python 3.13, pytest. **Date:** 2026-07-28.

1. Summary

Tests passed	325 / 325
Failures	0
Line coverage (<code>models.frequency</code>)	99.67%
Test modules	16

Every module the model added in Stages 1–6 carries its own suite; the loss, event-loss-table, rate-process and sensitivity modules are at 100% line coverage.

2. Coverage by test module

Module	Tests	Covers
<code>test_pot.py</code>	34	peaks-over-threshold extraction, declustering, threshold search
<code>test_weights.py</code>	38	population weights, catalogue coverage
<code>test_families.py</code>	30	Poisson / Negative-Binomial fit and calibrated selection
<code>test_ylt.py</code>	33	year-loss sampler and the closed-form reconciliation gate
<code>test_calibrate.py</code>	24	calibration orchestration, fallback, provenance
<code>test_events.py</code>	22	event aggregation, identity, hours-clause grouping
<code>test_rate_process.py</code>	20	constant / trend rate processes, term structure
<code>test_sensitivity.py</code>	20	rate, distributional and λ -uncertainty sensitivity
<code>test_losses.py</code>	19	loss-weighted YLT, AEP/OEP
<code>test_frame.py</code>	14	event frame, identifier resolution
<code>test_persist.py</code>	14	rate persistence and exact round-trip
<code>test_rate.py</code>	13	exceedance-rate estimation
<code>test_subject_losses.py</code>	13	loss adapter, coverage-scaling centralisation
<code>test_elt.py</code>	10	event loss table construction and export
<code>test_wind_lambda.py</code>	9	per-peril wind λ , decoupled-wind opt-in
<code>test_roundtrip.py</code>	5	serialisation golden-master

3. Key properties under test (self-tests, not just line coverage)

- **Reconciliation gate.** The Monte Carlo year simulation is checked against its exact closed form $1 - \exp(-\lambda p)$ on every run; tests pin that the deviation is within sampling standard errors and that the z-score statistic is itself calibrated (`test_ylt.py`).
- **Coverage-scaling trap closed.** The event loss table scales λ by catalogue coverage while the sampler does not; a test asserts the two paths agree so the units cannot drift (`test_subject_losses.py`).
- **Calibrated family selection.** The Negative-Binomial false-positive rate on genuine Poisson counts is held below the configured significance while real over-dispersion is still caught (`test_families.py`).
- **Behaviour-preserving seams.** The non-stationary term structure, decoupled wind and monetary-loss uplift each reduce to the prior behaviour by default, asserted directly (`test_rate_process.py`, `test_wind_lambda.py`, `test_subject_losses.py`).
- **Exact serialisation.** `rate_from_dict(rate_to_dict(r)) == r`, and a partial provenance record raises rather than defaulting (`test_persist.py`, `test_roundtrip.py`).

4. Scope and honesty

This is **unit and integration-wiring** evidence: it demonstrates that the code implements its specification and that the numerical machinery is internally exact. It is **not** back-testing or independent validation — the model's outputs are not compared against observed flood frequencies, because the catchment is synthetic and λ is unvalidated (see the model's limitations and the λ sensitivity analysis). Those remain outstanding (validation questions VQ-002, VQ-008, VQ-009).

Reproduce: `pytest tests/models/frequency/ --cov=models.frequency` in the `PhysicalRisk` repository.